

Real-Time Face Detection Using MATLAB

VIVEK PANCHABHAIYA

Object detection and tracking are important in many computer vision applications, including activity recognition, automotive safety and surveillance. Presented here is an object detection system that can detect not only a human face but also eyes and upper body.

Face detection is an easy and simple task for humans, but not so for

and video indexing systems.

This program can be used to detect a face, eyes and upper body on pressing the corresponding buttons. The program output screen is shown in Fig. 1.

Software program

This real-time face detection program is developed using MATLAB version R2012a. A graphic user interface (GUI) allows users to perform tasks interactively through controls like switches and sliders. You can easily create a GUI and run it in MATLAB or as a standalone application. The initial program output of this project is shown in Fig. 1.

Viola-Jones algorithm. There are different types of algorithms used in face detection. Here, we have used Viola-Jones algorithm for face detection with MATLAB program. This algorithm works in following steps:

1. Creates a detector object using Viola-Jones algorithm
2. Takes the image from the video
3. Detects features
4. Annotates the detected features

MATLAB functions. The program (testing.m) has many functions. Do not edit the functions as these are linkers and non-executable codes. First, you have to find the format supported by the camera and its device ID using the command given below (also shown in Fig. 2):

```
info = imaqhwinfo('winvideo')
```

After finding the device ID, you can change the device ID number in your source code.

The device ID here is {1}, so we have written '1' in the code, as men-

tioned below:

```
vid = videoinput('winvideo',1,'YUY2_640x480');
```

We also have other formats in MATLAB. You can check which format your camera supports by using the commands below (also shown in Fig. 3):

```
info.DeviceInfo(1)
info.DeviceInfo.SupportedFormats
```

In Fig. 3, you can see that format 'YUY2_160x120' is the one supported by the camera by default. But, there are other formats (resolutions) that your camera can support, as shown in the last line of this screenshot. If you select a different format and device number, you should make changes in the source code accordingly. To detect a face or a particular feature on the faces of people, use the following steps in MATLAB program (testing.m):

1. Define and set-up your cascade object detector using the constructor:

```
detector=vision.CascadeObjectDetector
```

It creates a system object detector that detects objects using Viola-Jones algorithm. Its classification model property controls the type of object to detect. By default, the detector is configured to detect faces.

2. Call the step method with input image I, cascade object detector, points PTS and any other optional properties.

Below is the syntax for using the step method. Use the step syntax with input image I, selected cascade object detector and other optional properties to perform detection.

```
BBOX = step(detector, I)
```

It returns BBOX, an M-by-4 matrix defining M-bounding boxes, containing detected objects. This method performs multi-scale object detection on input image I. Each row of output matrix BBOX contains a four-element vector (x, y, width and height) that specifies in pixels, the upper-left cor-



Fig. 1: Program output screen

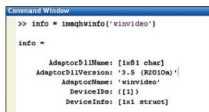


Fig. 2: Checking the device ID

computers. It has been regarded as the most complex and challenging problem in the field of computer vision due to large intra-class variations caused by the changes in facial appearance, lighting and expression. Such variations result in the face distribution to be highly nonlinear and complex in any space that is linear to the original image space. Face detection is the process of identifying one or more human faces in images or videos. It plays an important part in many biometric, security and surveillance systems, as well as image